Since, $\vec{F}.\vec{v} = 0 \Rightarrow \vec{F}$ and $\vec{v}$ are perpendicular to each other. The angle between Lorentz force and velocity of the charged particle is 90°. Thus Lorentz force changes the direction of the velocity but not the magnitude of the velocity. Hence Lorentz force does no work and also does not alter kinetic energy of the particle.

3.10.2 Motion of a charged particle in a uniform magnetic field

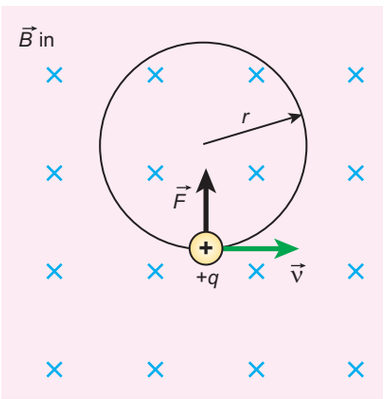


Figure 3.45 Circular motion of a charged particle in a perpendicular uniform magnetic field

Consider a charged particle of charge q having mass m entering into a region of uniform magnetic field $\vec{B}$ with velocity $\vec{v}$ such that velocity is perpendicular to the magnetic field. As soon as the particle enters into the field, Lorentz force acts on it in a direction perpendicular to both magnetic field $\vec{B}$ and velocity $\vec{v}$.

As a result, the charged particle moves in a circular orbit as shown in Figure 3.45. The Lorentz force on the charged particle is given by

$$\vec{F} = q(\vec{v} \times \vec{B})$$

Since Lorentz force alone acts on the particle, the magnitude of the net force on the particle is

$$\sum_i F_i = F_m = qvB$$

This Lorentz force acts as centripetal force for the particle causing it to execute circular motion. Therefore,

$$qvB = m\frac{v^2}{r}$$

The radius of the circular path is

$$r = \frac{mv}{qB} = \frac{p}{qB} \qquad (3.57)$$

where $p = mv$ is the magnitude of the linear momentum of the particle. Let T be the time taken by the particle to finish one complete circular motion, then

$$T = \frac{2\pi r}{v} \qquad (3.58)$$

Hence substituting (3.57) in (3.58), we get

$$T = \frac{2\pi m}{qB} \qquad (3.59)$$

Equation (3.59) is called the **cyclotron period**. The reciprocal of time period is the frequency f, which is

$$f = \frac{1}{T}$$

$$f = \frac{qB}{2\pi m} \qquad (3.60)$$

In terms of angular frequency ω,

$$\omega = 2\pi f = \frac{q}{m}B \qquad (3.61)$$

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